CMSC250 Quiz 1

Discrete Mathematics | Instructor Saurav Patel

UID (9 digits): | | | | | | | | | |

Student Name: _____

Directory ID: _____

Instructions: Answer all questions. Show your reasoning where requested. For multiple-choice questions, circle one answer.

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1. Truth Table (6 points). Complete the truth table for the proposition $(p \wedge \neg q) \vee q$. Use T for true and F for false.

p	q	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q) \vee q$
T	T			
T	F			
F	T			
F	F			

2. Multiple Choice (2 points). Which expression is logically equivalent to $\neg(p \vee q)$?

A. $\neg p \vee \neg q$

B. $\neg p \wedge \neg q$

C. $p \wedge q$

D. $p \vee \neg q$

3. Multiple Choice (2 points). Which statement is the contrapositive of "If n is divisible by 4, then n is even"?

A. If n is even, then n is divisible by 4.

B. If n is not divisible by 4, then n is not even.

C. If n is not even, then n is not divisible by 4.

D. If n is divisible by 4, then n is not even.

4. Short Answer (4 points). Define what it means for two propositions to be *logically equivalent*. Give one example of two logically equivalent propositions.

5. Integer Proof (6 points). Prove that for every integer n, the integer $n^2 + n$ is even. Clearly state the definition of an even integer and justify your steps.